Food Formulation

Science

May, 2026

Ollscoil Teicneolaíochta an Oirdheiscirt
South East Technological University

Food Formulation (A06335)

Short Title:	Food Formulation		
Faculty	Science and Computing		
Department	Science		
Cluster	Food Science		
Author	EDUGGAN		
Credits	5	**Level:**	Intermediate

Description of Module / Aims

The aim of the module is to introduce students to a variety of ingredients commonly used in the food industry. The functionality and applications of these ingredients will be assessed in terms of product development. The student should be able to understand the principles of food formulation.

Programmes

FOOD-0021	BSc (Hons) in Agricultural Science (WD_SAGRI_B)	3 / 5 / M
FOOD-0021	BSc (Hons) in Food Science and Innovation (WD_SFSIN_B)	3 / 6 / M
FOOD-0021	BSc in Food Science with Business (WD_SFDSC_D)	3 / 6 / M

Indicative Content

- Food proteins: properties and functionality
- Food lipids: properties, functionality; emulsion formation
- Food carbohydrates: properties and functionality
- Starter cultures for fermented food products
- Sensory aspects of food: colour, flavour and texture
- New food product development: probiotics, functional foods

Learning Outcomes

On successful completion of this module, a student will be able to:

1. Appraise the functions of a variety of food ingredients for use in food products.
2. Determine the rationale for using particular ingredients in food products.
3. Apply microbiology to food processing and especially in fermented food development and production.
4. Illustrate the application of sensory analysis in food development.
5. Differentiate different functional foods.
6. Investigate food ingredient functionality using standard laboratory techniques.
7. Integrate theoretical knowledge and practical skills to conduct food formulation on a given product.

Learning and Teaching Methods

- Lectures will provide the theory of food formulation and functionality.
- Practical classes will enable students to investigate food ingredient functionality, analyse results and prepare reports.
- A group project will encourage the students to apply the theory of food ingredient functionality in the manufacture of food products.

Assessment Methods

	Weighing	Outcomes Assessed
Continuous Assessment	100%	
Practical	50%	1-7
Practical	25%	6
Group Project	25%	7

Assessment Criteria

<40%: Little or no demonstration of an understanding of the fundamentals of the learning outcomes. Lack of competence with regards to use of associated laboratory skills, techniques and instrumentation.

40%-49%: Demonstrate limited understanding of the fundamentals of the learning outcomes. Will display some ability with regard to the use of associated laboratory skills, techniques and instrumentation.

50%-59%: Demonstrate satisfactory understanding of the fundamentals of the learning outcomes. Will display reasonable competence with regards to use of associated laboratory techniques instrumentation and good laboratory practice.

60%-69%: Demonstrate sound knowledge of the all topics contained in the learning outcomes and display good competence with regards to use of associated laboratory instrumentation and very good laboratory practice.

70%-100%: Demonstrate comprehensive knowledge and understanding of all learning outcomes. Will display advanced competence and independence with regards to use of associated laboratory techniques and instrumentation and excellent laboratory practice.

Note:

- In addition to the normal regulations, a minimum of 35% must be achieved in both the final exam and continuous assessment components of the module. In cases where continuous assessment has a practical component attendance of at least 75% in the practicals is mandatory.

Learning Modes

Learning Mode	F/T Hours	P/T Hours
Lecture	24	
Practical	36	
Independent Learning	75	

Essential Material

- Kilcast, D. _Instrumental Assessment of Food Sensory Quality : A Practical Guide_. Palo Alto: Woodhead Publishing, 2013.
- Velisek, J. _The Chemistry of Food_. Chichester: Wiley-Blackwell, 2013.

Supplementary Material

- "ift.org." https://www.ift.org/
- Aguilera, H.M. and P.J. Lillford. _Food Materials Science: Principles and Practice_. New York: Springer, 2008.

- Moskowitz, H.R., J.H. Beckley and A.V.A. Resurreccion. _Sensory and consumer research in food product design and__development_. Ames, Iowa: Blackwell Publishing, 2012.

Requested Resources

- Room Type: Chemistry Lab